English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Software Engineering, Year 3, Groups 4/5, 2020/2021 Test 2

Your grade is 2 plus the number of points earned. Time allowed: 4 hours. Good luck.
Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. (2 points)

1. (0.5 points) Two cards are drawn from a standard 52-card deck. Let X be the number of aces and Y the number of hearts. Find the joint distribution and correlation of X, Y.

2. (1 point) Let X be an asset price and Y the average deposit interest rate (both daily and in dimensionless units). Their joint density is $f_{X,Y}(x,y) = c(x^2 + e^y x)$ for $0 < x, y < 1$, and zero elsewhere, where c is a constant. Find c and the covariance of X, Y. What is the expected asset price if the interest rate is 0.5?

3. (0.5 points) Let X have density $f_X(x) = \frac{1}{4} x e^{-x/2}$ for $x \geq 0$ and zero otherwise. Find the density of $Y = -2X + 2$.

Problem 2. (0.5 points)

According to official data, Sofia's historical average January temperature is $-0.5°C$. Let X be the average January temperature in Sofia in degrees Celsius, assumed normally distributed. The variance is usually estimated from data too, but here we consider a different (not recommended) method.

Suppose someone regards a temperature above $20°C$ as an extreme anomaly with probability 1 percent. What variance estimate does this imply? What is then the probability of a temperature below $-15°C$?

Problem 3. (1.5 points) A point A is chosen randomly in the disk $k(O, 1)$ with centre O and radius 1. Let $X = |OA|$. Can you guess the mode and median? Justify your guesses. What would you expect $\mathbb{E}X$ to be? *(The mode of a discrete distribution is a value with maximum probability. In the continuous case, analogously, we seek a value maximizing* f_X*. We call* a *a median of* X *if* $\mathbb{P}(X \leq a) = \mathbb{P}(X \geq a) = 1/2$*.)*

1. (0.75 points) Find the distribution function, density, expectation and variance of X.

2. (0.75 points) Now let A_1, A_2, A_3 be three points chosen randomly and independently in the same disk. What is the expected distance to the point nearest the centre? And to the farthest? *(Bonus: Find the expected distance to the middle point. Should it equal* $\mathbb{E}X$*?)*